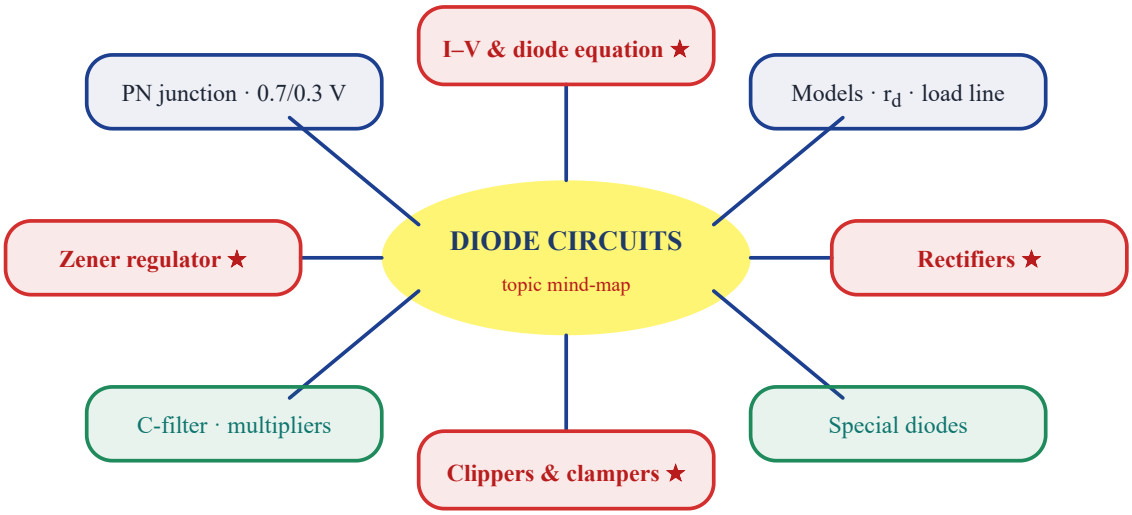
Diode Circuits

I–V · models · Zener · rectifiers · clippers · clampers: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = applied / circuits

Syllabus coverage

"Diode circuit: **current-voltage characteristics of a diode**, DC and AC models, dynamic resistance and capacitance, load line, **Zener regulator**, **half wave and full wave rectifier**, voltage multiplier, **clipper and clamper**". A one-page semiconductor primer comes first; transistors (BJT, FET, amplifiers) are the next lessons of this chapter.

High-yield list

① Si 0.7 V · Ge 0.3 V
② $r_d = 26\ mV/I_D$
③ HW: γ 1.21, η 40.6%
④ FW: γ 0.48, η 81.2%
⑤ PIV: CT $2V_m$, else V_m
⑥ ripple: f (HW), 2f (FW)
⑦ $I_Z = I_S - I_L$

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ Semiconductors & PN junction …… p. 2
✓ Diode equation, I–V, breakdown … p. 3
✓ Models, r_d, capacitance, load line … p. 4
✓ Zener regulator, special diodes … p. 5
✓ Half-wave & centre-tap rectifiers … p. 6
✓ Bridge rectifier & comparison …… p. 7
✓ Capacitor filter, multipliers …… p. 8
✓ Clippers & clampers …… p. 9
✓ Night-before sheet …… p. 10
✓ Practice MCQs + written …… p. 11

1 Semiconductors in One Page

Definition: A semiconductor has resistivity between a conductor and an insulator. Si and Ge are **group-IV (4 valence electrons)**, held in covalent bonds. Band gap: **Si 1.12 eV** · Ge 0.67 eV · GaAs 1.42 eV. Resistance **falls** as temperature rises (negative temperature coefficient).

Intrinsic = pure crystal: $n = p = n_i$ (Si ≈ 1.5×10^{10} cm^{-3} at 300 K).

Extrinsic = doped (≈ 1 impurity atom per 10^6–10^8 Si atoms) → conductivity rises sharply.

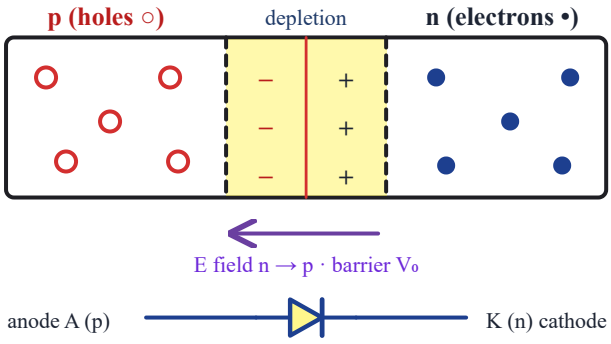
Mass-action law: $\mathbf{n \cdot p = n_i^2}$

	n-type	p-type
Impurity	pentavalent (donor)	trivalent (acceptor)
Examples	P, As, Sb	B, Al, Ga, In
Majority	electrons	holes
Minority	holes	electrons
Fixed ions	+ donor ions	− acceptor ions

Memory trick: **Five gives** a spare electron → n-type ("PAS it on": P, As, Sb). **Three takes** one → a hole → p-type (B, Al, Ga, In).

Exam trap: An n-type crystal is electrically **neutral**, not negative: every free electron left behind a fixed + donor ion.

2 The PN Junction



Diffusion of carriers across the junction leaves **immobile ions** → a **depletion (space-charge) region**, ≈ 0.5 µm wide.

Its field creates the **barrier potential** V_0: **Si 0.7 V Ge 0.3 V** (25 °C).

V_0 falls ≈ **2 mV per °C** rise.

	Forward bias (p to +)	Reverse bias (p to −)
Depletion	barrier lowered, region **narrows**	barrier raised, region **widens**
Current	large (mA–A), majority carriers	tiny I_S (Si nA, Ge µA), minority carriers
Behaves like	closed switch (after 0.7 V)	open switch (until breakdown)

Reverse saturation current I_S roughly **doubles for every 10 °C** rise. A BJT (next lesson) is simply two PN junctions back to back.

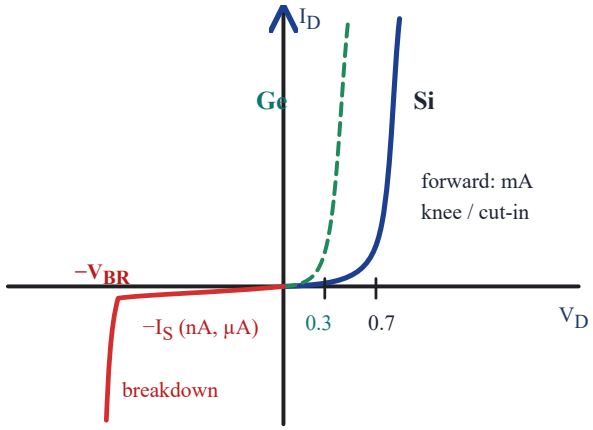
P to Positive = Forward

3 Shockley Diode Equation

Equation: $\mathbf{I_D = I_S\,(e^{V_D/\eta V_T} - 1)}$ $V_T = kT/q \approx$ **26 mV at 300 K**

I_S = reverse saturation current · η = ideality factor: 1 (Ge, IC silicon) to 2 (discrete Si at low current)

- Forward, $V_D \gg V_T$: $I_D \approx I_S\, e^{V_D/\eta V_T}$ (exponential)
- Reverse, $V_D \ll -V_T$: $I_D \approx \mathbf{-I_S}$ (flat, tiny)
- Cut-in (knee) voltage: **Si 0.7 V, Ge 0.3 V**: below it the current is negligible
- Current ×10 needs only $\Delta V = \eta V_T \ln 10 \approx$ **60 mV** (η = 1)
- V_D falls ≈ 2 mV/°C at constant current: diodes work as temperature sensors



I–V curve (reverse scale exaggerated)

Worked example ✍️: I_S = 10 fA, η = 1, V_T = 26 mV. Find V_D at 1 mA and at 10 mA.

$V_D = \eta V_T \ln(I_D/I_S) = 0.026 \ln(10^{11}) =$ **0.659 V** at 1 mA; at 10 mA: $0.026 \ln(10^{12}) =$ **0.718 V** (+60 mV per decade). This is why "0.7 V" works for any normal current.

Exam trap: I_S comes from **minority carriers**, so it depends on **temperature**, not on the reverse voltage. Ge has a far larger I_S than Si: one reason Si replaced Ge.

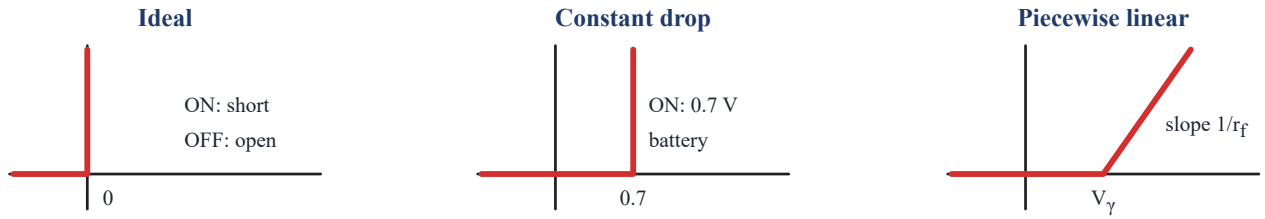
4 Reverse Breakdown: Zener vs Avalanche

	Zener breakdown	Avalanche breakdown
Doping	heavy doping, very thin layer	light doping, wide layer
Mechanism	field emission: strong field (~10^7 V/m) pulls electrons out of bonds	impact ionisation: carriers multiply by collisions
Voltage	below ≈ **5 V**	above ≈ 6 V (up to kV)
Temp. coeff.	**negative** (V_Z falls as T rises)	**positive**

Memory trick: **Z**ener = **Z**apped by the field (thin, low V, TC −). **A**valanche = **a** snowball of collisions (wide, high V, TC +). At 5–6 V the two cancel → TC ≈ 0.

Practical link: Breakdown is not destructive; heat (P = VI) is. 1N4001…1N4007: 1 A, PIV 50…**1000 V**; 1N4148: fast switching diode.

5 DC Models of a Diode



Model	ON (forward)	OFF	Use when
Ideal	short, $V_D = 0$	open	$V_S \gg 0.7$ V; quick check
Constant drop	V_D = **0.7 V** (Ge 0.3 V)	open	default in exam numericals
Piecewise linear	$V_\gamma + r_f$ in series	open	r_f given

> **Worked example ✍️:** 5 V source, 1 kΩ, Si diode forward-biased. Ideal: I = 5/1k = 5 mA · constant drop: (5 − 0.7)/1k = **4.3 mA** · PWL (r_f = 20 Ω): 4.3/1020 = 4.22 mA.

6 Dynamic Resistance & Capacitance (AC model)

Static (DC): $R_D = V_D/I_D$ = 0.7 V/4.3 mA = 163 Ω. **Dynamic (AC):** $\mathbf{r_d = \eta V_T/I_D \approx 26\ mV/I_D}$ = slope at Q; at 4.3 mA r_d = **6.0 Ω** (+ bulk $r_B \approx$ 0.1–2 Ω).

Capacitance	Dominant in	Behaviour
Transition (depletion) $C_T = \varepsilon A/W$	reverse bias	falls as V_R rises (W wider) → **varactor**
Diffusion (storage) C_D	forward bias	$\propto I_D$; stored charge → t_{rr}

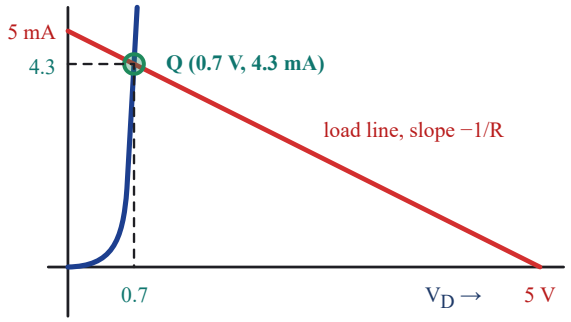
> **Exam trap:** r_d **falls** as current rises (26 Ω at 1 mA, 2.6 Ω at 10 mA); small signals see r_d, never R_D.

7 Load Line & Q-Point

KVL: $\mathbf{V_S = V_D + I_D R}$ → a straight line on the I–V plane.

1. $I_D = 0 \rightarrow V_D = V_S$ (x-intercept)
2. $V_D = 0 \rightarrow I_D = V_S/R$ (y-intercept)
3. Q-point = where the line cuts the diode curve

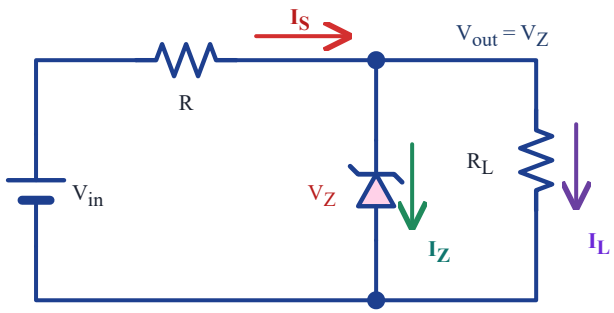
Larger R → flatter line.



V_S = 5 V, R = 1 kΩ

8 Zener Diode & Shunt Regulator

Definition: Heavily doped diode made for **reverse breakdown**: V_Z stays constant from I_{ZK} to I_{ZM}. In a regulator: **reverse-biased, in parallel with the load**.



Formulas:

$I_S = (V_{in} - V_Z)/R$

$I_L = V_Z/R_L$

$\mathbf{I_Z = I_S - I_L}$

$P_Z = V_Z I_Z \cdot I_{ZM} = P_{ZM}/V_Z$

Zener ON only if $V_{in}R_L/(R + R_L) \geq V_Z$

Load reg. $= (V_{NL} - V_{FL})/V_{FL} \times 100\%$

Worked example ✍️: $V_{in} = 20$ V, $V_Z = 12$ V, R = 400 Ω.

1. No load: $I_Z = I_S = (20 - 12)/400 =$ **20 mA** (worst case: $P_Z = 240$ mW)
2. $R_L = 1.2$ kΩ: $I_L = 12/1.2k = 10$ mA → $I_Z = 20 - 10 = 10$ mA, $P_Z = 120$ mW
3. Taking $I_{ZK} \approx 0$: $I_{L(max)} = 20$ mA → $R_{L(min)} = 12/0.02 =$ **600 Ω**

Exam trap: A **forward-biased** Zener is just a 0.7 V diode; it regulates only in reverse.

9 Special-Purpose Diodes



Diode	Bias	Key idea	Used in
LED	forward	emits light (GaAs IR, InGaN blue)	indicators
Photodiode	reverse	reverse current ∝ light intensity	light sensors
Varactor	reverse	C_T falls as V_R rises	tuning, VCO
Schottky	forward	metal–semiconductor, $\mathbf{V_F}$ **0.2–0.4 V**, no t_{rr}	SMPS, 74LS
Tunnel	forward	very heavy doping → negative resistance	microwave osc.

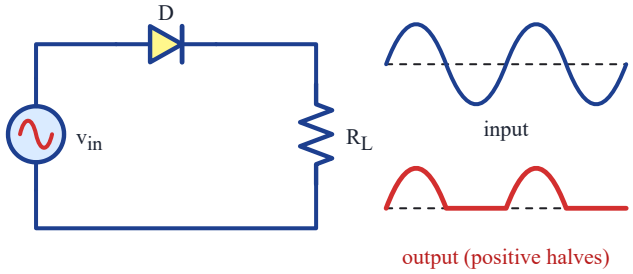
Memory trick: Arrows **out** = LED, arrows **in** = photodiode. Reverse-biased trio: **Zener, photodiode, varactor**.

LED ✍️: 5 V, $V_F = 2$ V, 15 mA → R = 3/0.015 = **200 Ω**.



linear DC power supply: every block of it is in this lesson

10 Half-Wave Rectifier (HWR)



HWR results:

$V_{dc} = \mathbf{V_m/\pi = 0.318\ V_m}$ · $V_{rms} = V_m/2$

Ripple factor γ = **1.21** · ripple frequency = **f**

$\eta_{max} = 4/\pi^2 \approx$ **40.6%** · PIV = $\mathbf{V_m}$

Form factor 1.57 · peak factor 2 · TUF 0.287

Derivation: $V_{dc} = (1/2\pi)\int_0^{\pi} V_m \sin\theta\, d\theta = V_m/\pi$

$\gamma = \sqrt{((V_{rms}/V_{dc})^2 - 1)} = \sqrt{(\pi^2/4 - 1)} = 1.21$

$\eta = P_{dc}/P_{ac} = (V_m/\pi)^2/(V_m/2)^2 = 4/\pi^2 = 0.405$ (books print 40.6%)

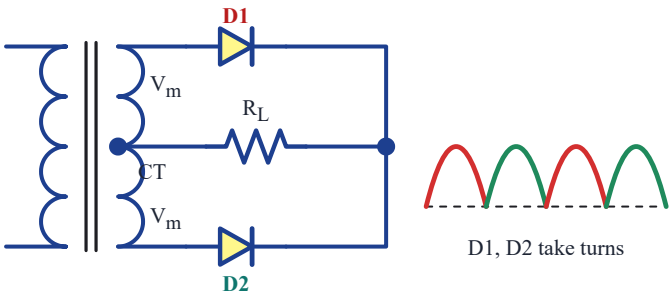
Worked example ✍️: 230 V, 50 Hz mains, 10 : 1 transformer, Si diode.

$V_s = 23$ V rms → $V_m = 23\sqrt{2} = 32.5$ V

$V_{dc} = (32.5 - 0.7)/\pi =$ **10.1 V** (ideal 10.4 V)

PIV = 32.5 V · ripple at **50 Hz**

11 Centre-Tap Full-Wave Rectifier



CT results:

2 diodes + centre-tapped transformer

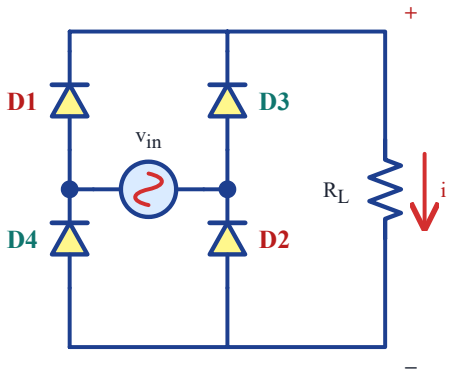
$V_{dc} = 2V_m/\pi = 0.637\ V_m$

$\gamma = 0.48$ · $\eta_{max} = 81.2\%$ · ripple 2f

PIV = $\mathbf{2V_m}$ · TUF 0.693

Exam trap: In the CT circuit V_m is the peak of **each half** of the secondary. The OFF diode sees both halves in series → PIV = $2V_m$ (a frequently tested question).

12 Bridge Full-Wave Rectifier



red pair conducts in the + half, green pair in the − half

+ half (left node +): **D1 → R_L → D2**

− half (right node +): **D3 → R_L → D4**

Load current is **the same direction in both halves**.

Two diodes in every path → drop **1.4 V** (Si).

No centre tap; PIV only **V_m**; TUF 0.812.

The big comparison table (memorise)

Parameter	Half-wave	Centre-tap FW	Bridge FW
Diodes	1	2	4
V_{dc} (ideal)	$V_m/\pi = 0.318\ V_m$	$2V_m/\pi = 0.637\ V_m$	$2V_m/\pi = 0.637\ V_m$
V_{rms}	$V_m/2$	$V_m/\sqrt{2}$	$V_m/\sqrt{2}$
Ripple factor γ	**1.21**	0.48	0.48
Max efficiency η	**40.6%**	81.2%	81.2%
PIV per diode	V_m	**$2V_m$**	V_m
Ripple frequency	f (50 Hz)	2f (100 Hz)	2f (100 Hz)
Form factor · peak factor	1.57 · 2	1.11 · 1.414	1.11 · 1.414
TUF	0.287	0.693	0.812
Si drop in path	0.7 V	0.7 V	1.4 V

Memory trick: **Double the waves, double the DC and the efficiency**: 0.318 → 0.637, 40.6 → 81.2. Ripple "**one-two-one**" (1.21) vs "**four-eight**" (0.48). PIV: the **centre tap is the odd one out** ($2V_m$).

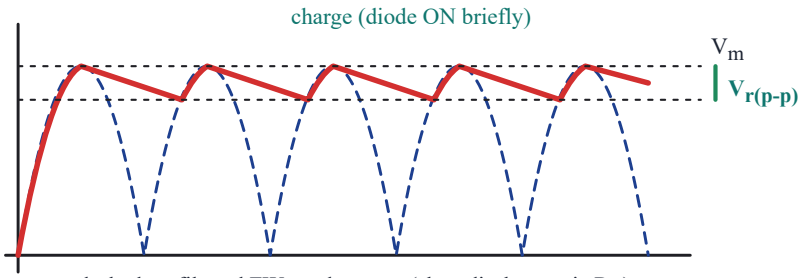
Exam trap: HWR γ = 1.21 > 1: the AC ripple is **larger** than the DC. The exact values are $4/\pi^2$ = 40.5% and $8/\pi^2$ = 81.1%; MCQ options use **40.6%** and **81.2%**.

Worked example ✍️: Bridge, 12 V rms secondary, Si diodes, R_L = 100 Ω, 50 Hz.

1. $V_m = 12\sqrt{2} = 16.97$ V; load peak = 16.97 − 1.4 = 15.57 V
2. $V_{dc} = 2 \times 15.57/\pi =$ **9.91 V** (ideal 10.8 V) → I_{dc} = 99 mA
3. PIV ≈ V_m = 17 V → a 1N4001 (50 V, 1 A) is enough; ripple at 100 Hz

13 Capacitor (Shunt) Filter

Idea: A large C across R_L charges to $\mathbf{V_m}$ at each peak and discharges slowly through R_L → DC with a small **sawtooth ripple**.



dashed: unfiltered FW · red: output (slow discharge via R_L)

Formulas: (f_r = ripple frequency)

$V_{r(p-p)} \approx I_{dc}/(f_r C) = V_m/(f_r R_L C)$

$V_{dc} \approx V_m - V_{r(p-p)}/2$

$\boldsymbol{\gamma = 1/(2\sqrt{3}\, f_r R_L C)}$

HW: $f_r = f$ · FW: $f_r = 2f$ → $\gamma = 1/(4\sqrt{3}\, f R_L C)$

Worked example ✍️: FW, 50 Hz, V_m = 15 V, R_L = 500 Ω, C = 1000 µF.

$I_{dc} \approx 15/500 = 30$ mA

$V_{r(p-p)} = 0.03/(100 \times 0.001) = \mathbf{0.30\ V}$

$V_{dc} \approx 15 - 0.15 = \mathbf{14.85\ V}$

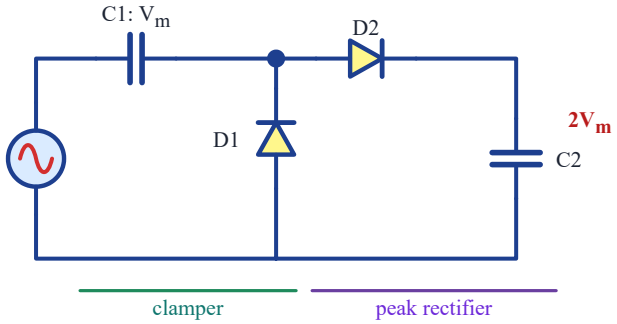
$\gamma = 1/(4\sqrt{3} \times 50 \times 500 \times 10^{-3}) \approx \mathbf{0.0058}$

Exam trap: **Half-wave + capacitor** → PIV = $\mathbf{2V_m}$ (C holds $+V_m$ while the input is at $-V_m$).

More C or bigger R_L → less ripple.

Other filters: series choke (heavy loads), LC (L-section), CLC **π-filter** (least ripple).

14 Voltage Multipliers



half-wave voltage doubler

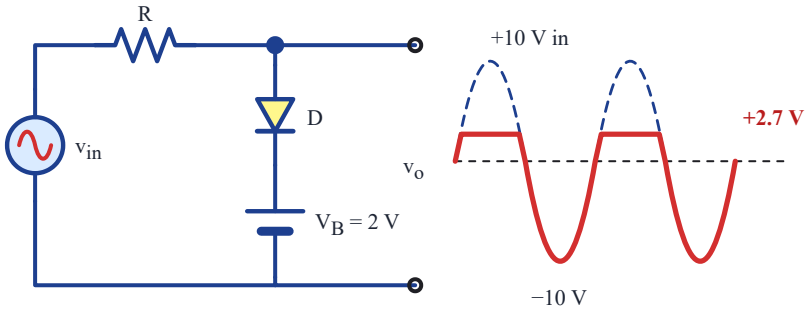
Tripler $\mathbf{3V_m}$ · quadrupler $\mathbf{4V_m}$ · n stages nV_m (Cockcroft–Walton). Small load currents only. Uses: CRT, X-ray, photocopier, laser printer.

1. − half: D1 ON, C1 charges to V_m
2. + half: source V_m + C1 V_m = $2V_m$; D2 ON, C2 charges to $\mathbf{2V_m}$
3. Each diode PIV = $2V_m$; C2 rated $2V_m$, C1 rated V_m
4. Full-wave doubler: both C rated V_m, ripple 2f

Memory trick: Doubler = **clamper + peak rectifier**. Each extra diode–capacitor pair adds one more V_m.

15 Clippers (Limiters)

Definition: Diode + resistor (no capacitor) that **removes the part of a waveform** above and/or below a level. **Series**: diode in series with the load; **shunt**: diode across it; **biased**: a battery sets the level.



biased shunt clipper: top cut at $V_B + 0.7$ V

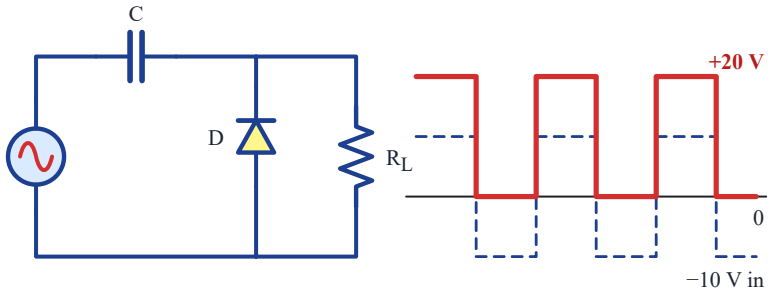
1. Find the input level where $V_{AK} = 0.7$ V: here $v_{in} = V_B + 0.7 =$ **2.7 V**
2. Above it: D ON → v_o held at 2.7 V
3. Below it: D OFF → $v_o = v_{in}$ (to −10 V)

Double (two-level) clipper: two biased diodes in opposite directions → output limited between $+(V_1 + 0.7)$ and $-(V_2 + 0.7)$.

Back-to-back Zeners: $\pm(V_Z + 0.7)$.

Exam trap: In a **series** clipper the conducting half comes out as $v_{in} - 0.7$ V, so a 10 V peak gives a **9.3 V** output peak, not 10 V. Uses: protecting inputs, limiters in radio/TV, noise removal.

16 Clampers (DC Restorers)



positive clamper: ±10 V in → 0 to +20 V out

C + diode (+ R_L) shift the **whole waveform** up or down. Shape and **peak-to-peak stay the same**.

Needs $\mathbf{R_L C \gg T/2}$ so C hardly discharges.

Worked example ✍️: ±10 V square wave, 1 kHz, C = 1 µF, R_L = 100 kΩ: τ = 100 ms ≫ T/2 = 0.5 ms ✓. Positive clamper, Si: **−0.7 to +19.3 V**; negative clamper: −19.3 to +0.7 V. p-p = 20 V in both.

Memory trick: Clipper **cuts**, clamper **climbs** (shifts). Clamper has a **C**apacitor. Draw the diode from output to ground: the waveform moves the way its **arrow points** (arrow up → positive clamper).

Night-Before-the-Exam Sheet

Fact sheet:

Si: E_g 1.12 eV, V_0 0.7 V · Ge: 0.67 eV, 0.3 V

n-type: P, As, Sb · p-type: B, Al, Ga, In

$I = I_S(e^{V/\eta V_T} - 1)$ · $V_T \approx 26$ mV

$r_d = \eta V_T / I_D \approx 26\text{ mV}/I_D$

60 mV per decade of current ($\eta = 1$)

V_D −2 mV/°C · I_S ×2 per 10 °C

Zener < 5 V (TC −) · avalanche > 6 V (TC +)

Load line: $V_S = V_D + I_D R$

Zener: $I_Z = (V_{in} - V_Z)/R - V_Z/R_L$

HW: V_m/π, γ 1.21, η 40.6%, PIV V_m

FW: $2V_m/\pi$, γ 0.48, η 81.2%

PIV: CT $2V_m$ · bridge V_m · HW + C $2V_m$

Ripple f: HW f · FW 2f

C-filter: $\gamma = 1/(4\sqrt{3}\, f R_L C)$ (FW)

Doubler $2V_m$ · tripler $3V_m$ · n stages nV_m

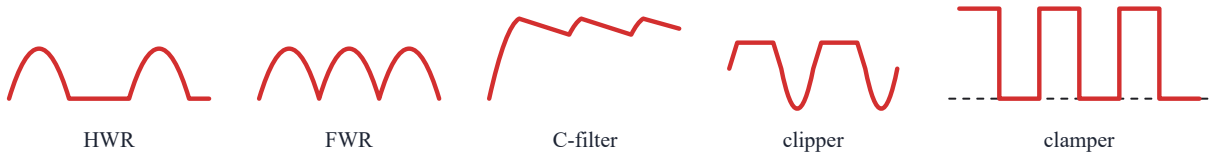
Clamper: $R_L C \gg T/2$, p-p unchanged

Top 12 one-liners for MCQs

1. Diode conducts anode → cathode (p → n)
2. Depletion region widens in reverse bias
3. Reverse current = minority carriers
4. Zener regulator: reverse-biased, ∥ load
5. Varactor: voltage-variable capacitance
6. Schottky: lowest V_F, fastest switch
7. LED forward-biased, photodiode reverse
8. FW ripple at 100 Hz on 50 Hz mains
9. Bridge: 4 diodes, no centre tap
10. FF: 1.57 (HW), 1.11 (FW)
11. TUF: 0.287 · 0.693 · 0.812
12. Clamper = DC restorer, clipper = limiter

Top traps: n-type is neutral · I_S depends on temperature, not V_R · $r_d \neq R_D$ · CT V_m is per half-winding · HWR $\gamma >$ 1 · HW + capacitor → PIV $2V_m$ · bridge loses 1.4 V · series clipper output peak = V_m − 0.7 · forward-biased Zener = 0.7 V diode

All tricks together: Five gives (n), three takes (p) · P to Positive = Forward · Zener zapped by field, avalanche snowballs · arrows out = LED, in = photodiode · double the waves, double DC and η · ripple one-two-one vs four-eight · centre tap is the odd PIV · clipper cuts, clamper climbs · doubler = clamper + peak rectifier



five output shapes to sketch from memory

✓ Practice MCQs (exam pattern, 10 marks)

1. Barrier potential of Ge at 25 °C:
(a) 0.7 V (b) 0.3 V
(c) 1.1 V (d) 0.1 V

2. Pentavalent doping of Si gives:
(a) p-type (b) intrinsic Si
(c) n-type (d) an insulator

3. r_d at I_D = 2 mA (V_T = 26 mV, η = 1):
(a) 52 Ω (b) 26 Ω
(c) 6.5 Ω (d) 13 Ω

4. PIV per diode in a bridge:
(a) V_m (b) $2V_m$
(c) $V_m/2$ (d) $\sqrt{2}\,V_m$

5. HWR ripple frequency, 50 Hz mains:
(a) 100 Hz (b) 25 Hz
(c) 50 Hz (d) 200 Hz

6. Maximum efficiency of a FW rectifier:
(a) 40.6% (b) 81.2%
(c) 100% (d) 121%

7. Diode used for electronic tuning:
(a) LED (b) Schottky
(c) photodiode (d) varactor

8. Ideal positive clamper, ±5 V in. Output:
(a) 0 to +10 V (b) −10 to 0 V
(c) −5 to +5 V (d) 0 to +5 V

9. TC of avalanche breakdown:
(a) negative (b) positive
(c) zero (d) infinite

10. V_{in} 15, V_Z 10 V, R 250 Ω, R_L 1k: I_Z?
(a) 20 mA (b) 30 mA
(c) 10 mA (d) 5 mA

Answer key → 1-b · 2-c · 3-d · 4-a · 5-c · 6-b · 7-d · 8-a · 9-b · 10-c

Written question: Draw a half-wave rectifier with its waveforms. Derive V_{dc} and V_{rms}, show that γ = 1.21 and $\eta_{max} \approx 40.6\%$, and find V_{dc} for a 12 V rms secondary with a Si diode. 8 marks

1. D conducts for $0 < \theta < \pi$: $v_o = V_m \sin\theta$; $v_o = 0$ for $\pi < \theta < 2\pi$ — 1
2. $V_{dc} = (1/2\pi)\int_0^{\pi} V_m \sin\theta\, d\theta = (V_m/2\pi)[-\cos\theta]_0^{\pi} = \mathbf{V_m/\pi}$ — 2
3. $V_{rms} = \sqrt{[(1/2\pi)\int_0^{\pi} V_m^2 \sin^2\theta\, d\theta]} = \sqrt{(V_m^2/4)} = \mathbf{V_m/2}$ — 2
4. $\gamma = \sqrt{((V_{rms}/V_{dc})^2 - 1)} = \sqrt{(\pi^2/4 - 1)} = \mathbf{1.21}$; $\eta = (V_m/\pi)^2/(V_m/2)^2 = 4/\pi^2 \approx \mathbf{0.406}$ — 2
5. $V_m = 12\sqrt{2} = 16.97$ V; with 0.7 V drop: $V_{dc} = 16.27/\pi = \mathbf{5.18\ V}$ — 1

Before the exam ✓

- ✓ Draw HW, CT, bridge + waveforms
- ✓ Comparison table row by row
- ✓ Zener: I_Z and $R_{L(min)}$
- ✓ Clipper/clamper outputs with 0.7 V

Also practise (written):
① Bridge: derive η = 81.2%
② C-filter ripple for given C
③ Double clipper sketch
④ Voltage doubler operation